

Green-Predict-Pro: Smart Agricultural Production Optimization Engine

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	—

Project Description

Green-Predict-Pro is a data-driven agricultural decision-support system that recommends the most suitable crop for a given plot of land based on soil nutrients (Nitrogen, Phosphorous, Potassium) and environmental parameters (temperature, humidity, pH, rainfall). The application is built on Flask and uses supervised machine-learning classifiers trained on the public crop-recommendation dataset. It empowers farmers to maximize yield, improve resource efficiency and adopt sustainable agricultural practices, while also providing an analytical view for researchers and policymakers.

Scenarios

Scenario 1 – Smart Crop Recommendation for Farmers: A farmer enters soil and environmental details (N, P, K, temperature, humidity, pH, rainfall). The system analyzes the data and recommends the most suitable crop for maximum yield and better farming efficiency.

Scenario 2 – Crop Suitability and Environmental Assessment: A user wants to check whether current soil and climate conditions are suitable for a specific crop. The application evaluates the input parameters and returns compatibility, suitability and productivity insights.

Scenario 3 – Agricultural Research and Policy Planning: A researcher or policymaker uses the platform to analyze crop-environment relationships and identify patterns for evidence-based, sustainable farming strategies.

Technical Architecture

Three-tier architecture — Presentation layer (HTML/CSS/JS + Bootstrap) → Application layer (Flask backend + scikit-learn ML model) → Data layer (Crop dataset CSV, SQLite/MySQL for users & history, joblib model artifacts).

(Note: An ER Diagram is included in the appendix showing users, predictions and crop_master tables.)

Pre-requisites

- Python 3.x programming proficiency
- Flask framework knowledge
- Familiarity with NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, SciPy

- HTML, CSS, JavaScript skills
- Git version control
- IDE such as VS Code / PyCharm / Jupyter Notebook

Project Workflow

Milestone 1: Data Collection & Understanding

- Activity 1.1: Download the crop-recommendation dataset from Kaggle.
- Activity 1.2: Inspect columns (N, P, K, temperature, humidity, ph, rainfall, label).
- Activity 1.3: Perform initial data quality checks (missing values, outliers).

Milestone 2: Exploratory Data Analysis

- Activity 2.1: Univariate & bivariate analysis using Pandas and Seaborn.
- Activity 2.2: Correlation heatmap of soil & climate parameters.
- Activity 2.3: Class-wise distribution of crops.

Milestone 3: Model Building

- Activity 3.1: Train Random Forest, Decision Tree, SVM and KNN classifiers.
- Activity 3.2: Compare accuracy, precision, recall and F1-score.
- Activity 3.3: Persist the best model and scaler as .pkl files.

Milestone 4: Flask Application Development

- Activity 4.1: Design the crop-input form (index.html).
- Activity 4.2: Build the /predict route which loads the model and returns the recommended crop.
- Activity 4.3: Build the /suitability route for the crop-suitability scenario.

Milestone 5: Frontend & Dashboard

- Activity 5.1: Style pages with Bootstrap for mobile responsiveness.
- Activity 5.2: Build a researcher dashboard with charts (Matplotlib / Chart.js).

Milestone 6: Testing & Deployment

- Activity 6.1: Unit-test the model and validation utilities.
- Activity 6.2: Load-test with JMeter.

- Activity 6.3: Deploy to Render / Heroku from GitHub.

Milestone 7: Conclusion

- Green-Predict-Pro delivers a production-ready ML-powered advisory that supports farmers, researchers and policymakers with evidence-based crop decisions.

Exploring the Website's Web Pages

Home / Input Page

Provides the crop-recommendation form where users enter N, P, K, temperature, humidity, pH and rainfall.

Result Page

Displays the recommended crop with a suitability score and a downloadable PDF report.

Suitability Page

Lets users choose a specific crop and returns whether current conditions are suitable.

Dashboard

Interactive charts showing crop-environment relationships for researchers.

Conclusion

Green-Predict-Pro combines modern machine learning with a simple, farmer-friendly web interface to deliver personalized crop recommendations. It contributes to higher yields, efficient resource use and sustainable agricultural practices, while providing a research-grade analytics view for policymakers. Future work includes multilingual UI, Weather-API auto-fill, mobile app support and integration with satellite-derived soil-moisture data.